ODS: Test-Time Adaptation in the Presence of Open-World Data Shift

Zhi Zhou, Lan-Zhe Guo, Lin-Han Jia, Ding-Chu Zhang, Yu-Feng Li

National Key Laboratory for Novel Software Technology, Nanjing University, Nanjing 210023, China

Test-time adaptation is essential for long-lived ML systems

- Modern ML systems face non-i.i.d. streams: changing weather, devices, scenes, populations.
- Test-Time Adaptation (TTA) updates models online using only unlabeled test data.
- Robust TTA is critical for safety, reliability, and long-lived deployments.

Open-world shifts demand simultaneous covariate and label adaptation

- Real streams shift in both covariates $\mathcal{D}_t(X)$ and label distribution $\mathcal{D}_t(Y)$.
- Most TTA methods assume only covariate shift, ignoring changing class priors.
- Goal: Adapt online under simultaneous shifts without source data.

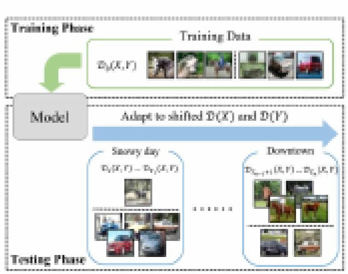
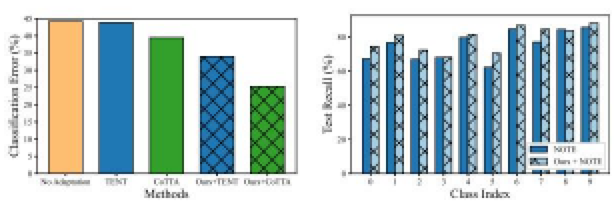


Figure: AODS示意：预训练模型在测试期自适应协变量与标签漂移

Label shift skews predictions and destabilizes standard TTA

- Changing class priors skews predictions and destabilizes self-training.
- Per-class recalls collapse for rare or surging classes.
- We must estimate and respect $\mathcal{D}_t(Y)$ during adaptation and prediction.



(a) Classification Error (%) (b) Test Recall(%)

Figure: CIFAR10标签移位：现有/鲁棒TTA难适配；本框架提升召回并增强TTA

Decoupling shifts enables plug-and-play open-world TTA

- Decouple shifts: track label distribution online and adapt features robustly.
- Plug-and-play framework ODS integrates with popular TTA (TENT, COTTA, NOTE).
- Theory links test error to label-distribution estimation and conditional error.

Tracking and optimizing label priors unlocks robust test-time predictions

- M^T: Distribution Tracker estimates current label distribution $\mathbf{w}_t$.
- M^O: Prediction Optimizer makes outputs consistent with $\mathbf{w}_t$.
- Base TTA focuses on covariate shift; ODS handles label shift.

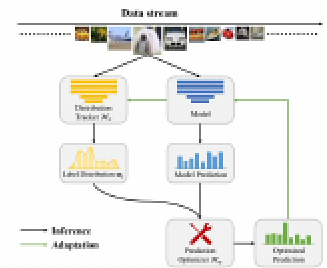


Figure: 两模块框架，估计并利用逐时间戳标签分布

Test error splits into prior mismatch and conditional error under GLS

- Test error gap decomposes into: label prior mismatch and conditional error.
- Key levers: minimize $\|\mathbf{1} - \mathcal{D}_t(Y)/\mathbf{w}_t\|_1$ and reduce Δ_{CE} via robust feature adaptation.
- Motivates accurate $\mathbf{w}_t$ tracking and $\mathbf{w}_t$-aware prediction optimization.

Δ_{CE} is the conditional error gap due to covariate shift; it is reduced by robust feature adaptation.

Confusion-based initialization stabilizes label-prior estimates

- Start with source-model confusion statistics (BBSE-style [BBSE, 2018]) to estimate $\mathbf{w}_t$.
- Use current batch predictions as weak signals of class prevalence.
- Provides a stable prior before refinement under changing covariates.

Feature-aware refinement improves label-prior accuracy online

- Optimize soft labels $\mathbf{z}_j$ with unsupervised objective: align with predictions, low entropy.
- Feature-similarity consistency using adapted model representations improves robustness.
- Compute batch $\mathbf{w}_t$ as mean of $\mathbf{z}_j$; smooth over time for stability.

Bias-aware reweighting protects minority classes during adaptation

- Reweight adaptation loss inversely to $\mathbf{w}_t$: $S(\mathbf{w}_t) = \mathrm{Normalize}(1 - \mathbf{w}_t)$.
- Mitigates skewed-batch bias and protects minority classes during updates.
- Robust across linear/convex/concave choices of $S(\mathbf{w})$.

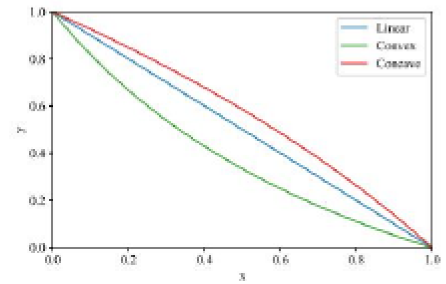


Figure: Illustration of different functions.

Prior-consistent prediction boosts robustness to label-shift noise

- Statistical plug-in: add $\log \mathbf{w}_t$ to logits (optimal if $\mathbf{w}_t$ exact; sensitive to noise).
- Conservative ensemble: geometric mean of model p and $\mathbf{z}_j$; more robust in practice.
- Apply M^O at inference using latest $\mathbf{w}_t$ for final predictions.

ODS is practical: light overhead, online, and source-free

- Runs per-batch online; lightweight temporal smoothing of $\mathbf{w}_t$ for stability.
- Conservative M^O adds modest runtime overhead while avoiding heavy augmentations.
- Operates on incoming unlabeled streams; no source data or labels needed during deployment.
- Compatible with diverse TTA backbones; minimal code intrusion.

We evaluate on CIFAR10-C/100-C as corruption benchmarks for TTA

- Corruption benchmarks: CIFAR10-C and CIFAR100-C for open-world image streams.
- Metrics: accuracy, label-prior estimation error, and runtime.
- Evaluate in-stream integration with strong TTA backbones under varying label-shift intensities.

ODS consistently augments strong TTA baselines in-stream

- Baselines considered:
  - SOURCE (no adaptation)
  - BN STATS [Ioffe & Szegedy, ICML 2015]
  - TENT [Wang et al., ICLR 2021]
  - EATA [Niu et al., NeurIPS 2022]
  - LAME [Boudiaf et al., NeurIPS 2022]
  - COTTA [Wang et al., CVPR 2022]
  - NOTE [Gong et al., ICCV 2023]
- ODS augments TENT/COTTA/NOTE with M^T and M^O in-stream.
- Consistent accuracy lifts across corruptions and timesteps; broad compatibility.

Results overview: ODS sets a new accuracy bar and sustains gains

- On CIFAR10-C, ODS+NOTE improves accuracy by about +3.5 points over NOTE (ICCV 2023); gains hold across most corruptions.
- Improvements persist from light to strong label-shift regimes with narrow variance.
- No overfitting to a particular shift intensity; boosts also seen for TENT (ICLR 2021) and COTTA (CVPR 2022).

ODS boosts diverse TTA methods across corruptions and time

- Consistent accuracy lifts across corruptions and timesteps.
- Works with entropy-minimization (TENT), augmentation-heavy (COTTA), and robust (NOTE) TTA.
- Demonstrates broad compatibility.

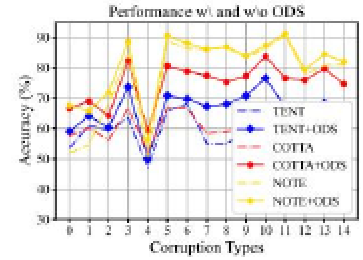


Figure: ODS framework effectively boosts TTA methods: TENT, COTTA, NOTE.

Stable gains across corruption types

- ODS maintains improvements under noise, blur, weather, digital corruptions.
- Gains persist over time in sequential streams.
- Confirms method generality in open-world settings.

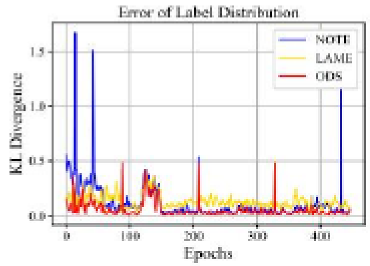


Figure: ODS framework effectively boosts TTA methods: TENT, COTTA, NOTE.

Accurate label-prior estimates drive better per-class balance

- ODS reduces over/under-prediction induced by shifted priors.
- Balancing via $\mathbf{w}_t$ improves minority and majority class recalls.
- Aligns predictions with real-time label frequencies.

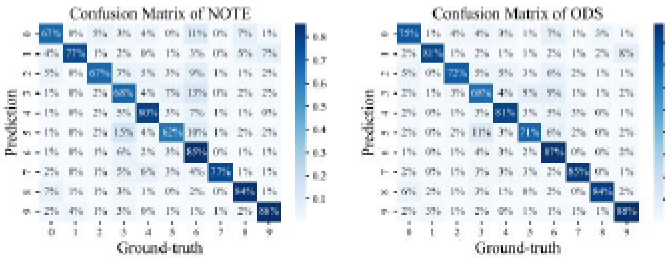


Figure: The confusion matrix of NOTE on CIFAR10 dataset.

Both MT and MO are necessary for peak performance

- M^T alone balances adaptation under skewed batches.
- Adding M^O yields the best overall gains across TTA backbones.
- Both modules contribute; together they yield the largest gains across TENT/COTTA/NOTE.

ODS remains stable across weighting shapes $S(w)$

- ODS is insensitive to the specific $S(w)$ shape (linear/convex/concave).
- Supports the design choice of simple linear default.
- Confirms stability of bias-aware adaptation.

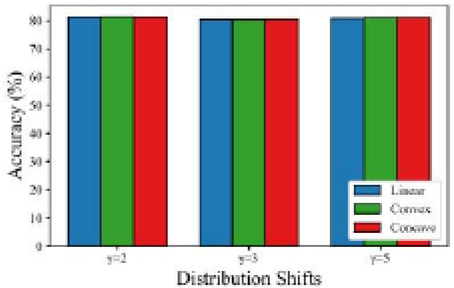


Figure: Illustration of different functions.

High-fidelity label-prior tracking reduces estimation error

- ODS sharply reduces divergence to the true $\mathcal{D}_t(Y)$ compared to baselines.
- Better $\mathbf{w}_t$ leads to improved per-class balance and accuracy.
- Validates the feature-aware refinement and temporal smoothing.

ODS has limits when label shift is absent or GLS is violated

- No-shift case: performance comparable but not always SOTA.
- Assumes generalized label shift; extreme domain drift may challenge $\mathbf{w}_t$.
- Future: confidence-aware M^O and mixed/unknown shift regimes.

Open-world TTA leaves key questions open

- When to apply prior-consistent prediction: detect/no-detect gating under weak or absent label shift.
- Beyond GLS: guarantees when $D_0(Z \mid Y) \neq D_t(Z \mid Y)$, partial overlap, or drifting conditionals.
- Scaling to many classes and rare events: stability with long-tailed streams and memory limits.
- Handling label-set changes and novel classes: graceful fallback and uncertainty calibration.
- Distributed/edge settings: communication-efficient tracking under privacy constraints.

ODS turns TTA into an open-world capability

- Problem: Simultaneous covariate and label shifts break standard TTA.
- Method: ODS decouples shifts via Distribution Tracker and Prediction Optimizer.
- Impact: Plug-and-play, theory-grounded, consistent gains across methods and shifts.

Broader impact and next steps extend ODS beyond vision

- Adaptive confidence-aware M^O and dynamic shift tracking theory.
- Extend to modalities beyond vision, incl. speech and VLMs.
- Enable reliable, continual ML in dynamic, real-world systems.

We cite all baselines used in this deck

- SOURCE: Source model without adaptation (standard supervised training).
- BN STATS: Ioffe and Szegedy. Batch Normalization. ICML, 2015.
- TENT: Wang et al. Tent: Fully test-time adaptation by entropy minimization. ICLR, 2021.
- EATA: Niu et al. Efficient test-time adaptation. NeurIPS, 2022.
- LAME: Boudiaf et al. LAME: Label-shift aware model adaptation. NeurIPS, 2022.
- COTTA: Wang et al. CoTTA: Continual test-time adaptation. CVPR, 2022.
- NOTE: Gong et al. NOTE: Robust test-time adaptation with neighborhood guidance. ICCV, 2023.

Your questions will sharpen the next steps

- Thank you for your attention!
- Questions and feedback are welcome.

We appreciate the community that enabled this work

- We thank collaborators, reviewers, and supporting institutions.
- Community resources and prior TTA methods enabled fair comparisons.
- Open-source tooling accelerated experimentation.